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Watching Responsibly: Toward Secure and Accountable AI Traffic Surveillance in Bangladesh

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***Abstract*—**Bangladesh's cities rank among the most congested in the world, with traffic in Dhaka often moving slower than walking pace. In 2026, the Dhaka Metropolitan Police deployed the country's first fully automated artificial intelligence traffic enforcement system, which uses computer vision and a deep-learning-based automatic number plate recognition (ANPR) pipeline to detect violations and file cases without human involvement. Early reports point to improved road discipline, and other cities, including Chattogram, are preparing to follow. This paper argues that the rollout has advanced faster than the safeguards needed to govern it responsibly. Because the system cross-references vehicle data with national identity and mobile records, it operates as a large surveillance infrastructure, yet it lacks clear data security standards and offers citizens little ability to contest automated decisions. Using the deployed Dhaka system as a case study, the paper evaluates it against two standards, security and accountability, and finds significant gaps in both. It then proposes a framework for responsible deployment built on three principles: secure by design, accountable by design and alignment with Bangladesh's emerging data protection law. As AI traffic surveillance spreads nationally, the paper makes the case for embedding these safeguards before the model becomes entrenched.

***Keywords*—**AI traffic surveillance, data security, algorithmic accountability, responsible AI, automated enforcement, Bangladesh

I. INTRODUCTION

Dhaka is among the most congested cities in the world; at peak hours its average traffic speed has been reported at roughly 4.8 kilometers per hour, slower than a brisk walk [1]. In May 2026, the Dhaka

Metropolitan Police (DMP) responded by launching the country's first fully automated artificial intelligence (AI) traffic enforcement system, which analyses existing closed-circuit television feeds to detect violations and file cases without human involvement [2], [3]. Early reporting describes improved lane discipline, and other cities, notably Chattogram, are preparing to follow [4].

This paper argues that the rollout has advanced faster than the governance needed to direct it: technology adoption is outpacing governance readiness. Because the system cross-references vehicle data with identity infrastructure, using vehicle registration and, where needed, national identity and mobile records to locate owners, it functions as more than traffic management. It is surveillance infrastructure. Yet its security standards are unstated and citizens have little means to contest an automated decision. The benefits are real and worth acknowledging: automated enforcement can improve discipline, reduce opportunities for petty roadside corruption and ease the manual policing burden. The argument here is not that the technology is harmful, but that security and rights protections must scale alongside deployment.

Three technologies underpin the system. Computer vision allows machines to interpret images and video; automatic number plate recognition (ANPR) detects a vehicle's plate and reads its characters using optical character recognition (OCR); and automated enforcement turns those readings into machine-generated penalties. Two governance concepts frame the analysis. Algorithmic accountability refers to the obligation to explain and answer for the decisions an automated system reaches [5]. Responsible AI is the broader, related commitment to building systems that stay lawful, ethical, and robust across their life cycle [6], [7].

Camera automation and ANPR are well established internationally, but where they are effectively governed they come with safeguards. The broader body of literature highlights that public-sector algorithmic systems present unique accountability, fairness and transparency issues not typically associated with traditional software [8], [9]. A recurring finding of surveillance research is that cameras built for one purpose are prone to function creep, so purpose limitation and oversight must be built in from the start [10], [11]. The pressing question for Bangladesh, then, is not whether the technology functions properly, but whether it is being deployed within a responsible framework.

This paper asks three questions: (1) how does Bangladesh's AI traffic enforcement system collect, link and protect citizen data; (2) what security and accountability gaps exist in its current deployment; and (3) how can responsible-AI principles guide a secure and accountable rollout as the system scales nationally? It contributes a Bangladesh-focused case study of a live deployment, a dual security-and-accountability framework for evaluating such systems and a responsible-deployment model for emerging AI enforcement systems in low-regulation settings.

II. METHODOLOGY

This study uses qualitative document and policy analysis rather than experimental benchmarking. No official technical documentation, procurement specification, or independent audit of the deployed system is in the public domain, so the system had to be reconstructed from external sources. The paper is therefore a conceptual security-and-governance evaluation rather than an empirical performance study.

The corpus has four strands. The first is contemporaneous news reporting on the deployment, drawn from the Dhaka Tribune, The Business Standard, the Global Voices Advox project, and Let's Data Science [1]-[4], [12]. The second is legal and regulatory material: the Personal Data Protection Ordinance of 2025 as amended in 2026, tracked through the Digital Policy Alert regulatory database [16], with the General Data Protection Regulation used as a comparator [18]. The third is peer-reviewed scholarship on algorithmic accountability, public-sector AI governance, and camera surveillance [5]-[11], [17]. The fourth is technical literature on automatic number plate recognition, comprising a recent review, a probabilistic benchmarking study, and locally developed research on Bangladeshi plates [13]-[15].

Two date windows apply. Deployment evidence covers May to June 2026, running from the launch of the system through the first reported prosecutions and the Chattogram proposal. Legal sources span 2016 to 2026, and the scholarly and technical literature spans 1999 to 2026, so that foundational surveillance research sits alongside current machine-learning work.

Claims were triangulated on a simple rule. A factual claim about the deployment was treated as confirmed only where two or more independent outlets reported it. Where a single source carries a claim, that source is named in the text so the reader can weigh it, and where a statement is the authors' interpretation rather than a reported fact, it is marked as such. Absence claims, such as the lack of a published false-positive rate, are stated as absence within this corpus rather than as proof that no such document exists. This three-tier scheme of confirmed, single-source, and inferred is applied throughout Section III.

III. RESULTS AND DISCUSSION

A. The Dhaka System: Anatomy and Case Study

As reported by the Dhaka Tribune [2], the system was introduced by the DMP in May 2026, using artificial intelligence at approximately 25 key intersections. The computer-vision-based systems automatically identify infractions based on analysis of existing camera feeds, marking them with alerts for violations like running a red light, crossing the stop line and blocking a lane as illustrated in Fig. 1. The system then reads the number plate, links the registration database of the Bangladesh Road Transport

Authority (BRTA) to identify the owner of the vehicle and sends an SMS prosecution notice, a sequence corroborated by [2] and [3]. In the absence of any owner, the authorities rely on BRTA records including mobile numbers and national identity (NID) data, as reported by [2]. This linkage rests on a single source, and it is the claim on which the surveillance argument of this paper depends. Cases are generated automatically, and the DMP has stated an aim of full automation within months, as reported by [3].

The salient point is the data linkage. A system presented as traffic management cross-references movement at fixed points with identity infrastructure. This combination, rather than any single camera, is what turns the deployment into surveillance infrastructure [10], [11], and it is the hinge on which the security and accountability questions turn. That characterisation is an inference the authors draw from the surveillance literature, not a claim made by the operator, and it is bounded to what the sources confirm: the system is reported to draw on identity and contact records to locate owners, which is a cross-referenced identity capability, not necessarily real-time mass tracking.

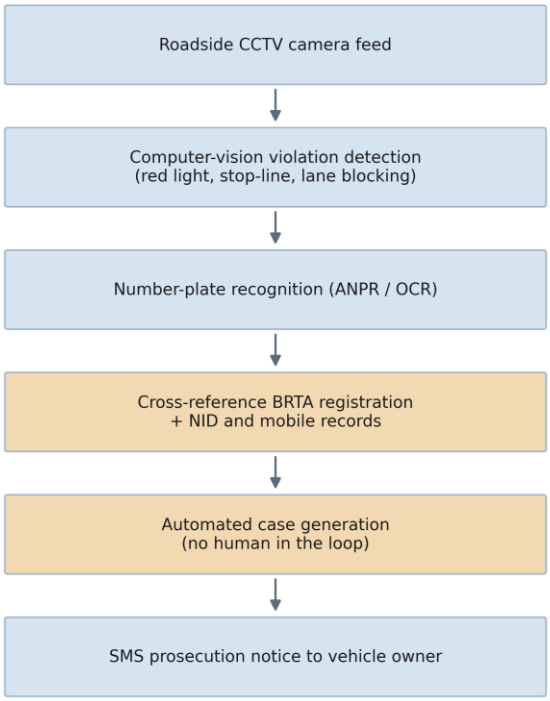


Fig. 1. End-to-end workflow of Dhaka's AI traffic enforcement, from camera capture and ANPR processing to BRTA and NID lookup and automated case generation. (Visual layout assisted by AI generation; conceptual pipeline design is original).

B. The Security Gap: Who Secures the Surveillance?

This is where a traffic tool becomes a security liability. A central system that ties vehicle movements to identity records is both a high-value dataset and a broad attack surface. Several risks follow. First, a breach of the central database would expose combined movement and identity data, a graver exposure than either alone. Second, insider misuse and unauthorised access are persistent risks wherever sensitive citizen data is concentrated without strong controls. Third, the camera endpoints and network links are themselves potential points of entry if not hardened. Fourth, the models can be evaded or manipulated: an early case reported by The Business Standard [12], in which a motorcyclist was jailed for taping over his plate to defeat the cameras, shows that the cat-and-mouse dynamic of adversarial evasion has already begun [12]. Fifth, and most consequential over time, is function creep, in which infrastructure built to enforce traffic law quietly widens into general movement monitoring unless its purpose is legally bounded [11].

None of this means the system should not exist. It means that security must be designed in, not bolted on. Established safeguards include encryption of data in transit and at rest, data minimization, role-based access control, zero-trust architecture, comprehensive audit logging, independent security audits and mandatory breach notification. Table I summarizes the principal gaps against these safeguards and frames the accountability gaps discussed in Section III-C. Coverage is listed separately as an equity gap, because excluding rickshaws is not a security or accountability failure but a question of who the system polices.

Table I. Security, accountability, and equity gaps in Dhaka's AI traffic system.

Area	Current situation	Risk	Proposed safeguard
Data storage	Central database cross-referenced with BRTA and NID records	Breach exposes movement and identity data together	Encryption, data minimization, access control
AI detection	Automated cases, no human review	False fines from plate or OCR errors	Human-in-the-loop review before prosecution
Equity (coverage)	Motor vehicles only; rickshaws excluded	Inequitable enforcement and blind spots	Inclusive detection and transparent scope policy
Appeals	No clear contestation pathway	Citizens cannot challenge errors	Public appeal portal and explainability
Oversight	Police-run, no independent audit	Function creep and unchecked surveillance	Independent oversight and published audits
Legal	Data law in force but with state exemptions; regulator not operational until about 2027	Weak enforceable protection	Close exemption loophole; operationalize data rights

C. The Accountability Gap: Justice without Recourse

The security gap concerns the data in the system; the accountability gap concerns the decisions it makes. The hallmark of the Dhaka system is that it is automatic, meaning that there is no human involvement [3]. Automated public-sector decisions raise well-documented accountability problems,

because those affected may be unable to understand or challenge a decision that no official actually made [5], [9].

Understanding why these errors occur requires a brief look at how the technology is built. Modern ANPR is a two-stage deep-learning pipeline: a convolutional neural network first detects the plate within the camera frame, and a second model reads its characters, both trained on large sets of labelled plate images [13]. Performance therefore depends heavily on how closely the training data match real deployment conditions. Models trained on clean, standardised plates tend to falter on the worn, dirty, non-standard, and sometimes hand-painted plates common in Bangladesh, a mismatch known as domain shift [14]. Local conditions sharpen this gap, since heavy monsoon rain, dust occlusion, and night-time glare from oncoming high beams all degrade a convolutional neural network's ability to localise and read a plate. Where data are thin or unrepresentative, accuracy falls and predictive uncertainty rises, which is precisely when wrongful cases become likely [15].

The technical reliability of the recognition stage is central here, and it keeps this a question about the AI and not only about policy. These degradation modes are well documented in the plate-recognition literature [13], [15]. Global Voices reports that the deployed system struggles with blurred or unclear plates [3]. Each failure mode can produce a false positive, and a false positive in this system is a wrongful prosecution. Coverage compounds the problem: the system targets motor vehicles and, as reported by [3], currently excludes rickshaws and other non-motorised vehicles, which is both a technical limit and a fairness problem, since enforcement falls unevenly across road users. Locally developed plate-recognition research confirms that building reliable ANPR for Bangladeshi plates is itself an open technical challenge [14].

Accountability requires recourse. Observers have noted that there is currently little public clarity on how a citizen contests an automated case, alongside a complete absence of published accuracy or false-positive metrics [1]. Both rest on a single source, and no such metrics appear in the corpus of Section II. There is a legal backdrop of a mix. Even though the Personal Data Protection Ordinance was created in 2025 and amended in 2026, making personal data a protected interest, commentators note that its protection is undermined by broad exemptions by the state from the obligations of the PDPO, and its enforcement structure is unlikely to be fully up and running until 2027 or so [16]. Mature regimes combine automated decisions with independent oversight, and require Data Protection Impact Assessments, and provide a right to contest, as Europe does under its framework [17], [18], and as long-running debates about camera surveillance in the United Kingdom and the Netherlands remind us [10], [11]. Bangladesh holds these rights on paper, but lacks the necessary recourse in practice.

D. Watching Responsibly: A Framework and Recommendations

The convergence in the AI-ethics literature around transparency, justice and fairness, non-maleficence, responsibility and privacy [7], together with the trustworthy-AI requirements of human oversight, robustness, privacy and data governance, transparency, fairness and accountability [6], can be translated into three concrete principles for AI traffic surveillance, shown in Fig. 2.

Secure by design: The safeguards in Section III-B should be built in from the start: encryption, data minimisation, access control, audit logging, and independent security audits, with the identity linkage protected as critical infrastructure.

Accountable by design: Prosecution should pass through human-in-the-loop review, in which the model flags a violation and a person validates it before a case is filed. Citizens should have access to an appeal portal and an explanation of the model, and the operator should publish accuracy metrics including precision, recall and false-positive rates, along with retention limits and an explicit purpose limitation to curb function creep.

Legal alignment: The rights citizens nominally hold under the data-protection ordinance should be made operational, the state-exemption loophole narrowed, and an independent body given oversight of audits and appeals.

From these principles follow concrete recommendations: an independent AI audit, a citizen appeal portal, a data-retention limit, public transparency reports, mandatory human oversight, and a privacy impact assessment before any expansion. Timing matters. Chattogram is adopting the model now and Sylhet has none yet, so safeguards can still be built in rather than retrofitted, which is invariably cheaper and more credible than reform after the fact [8].

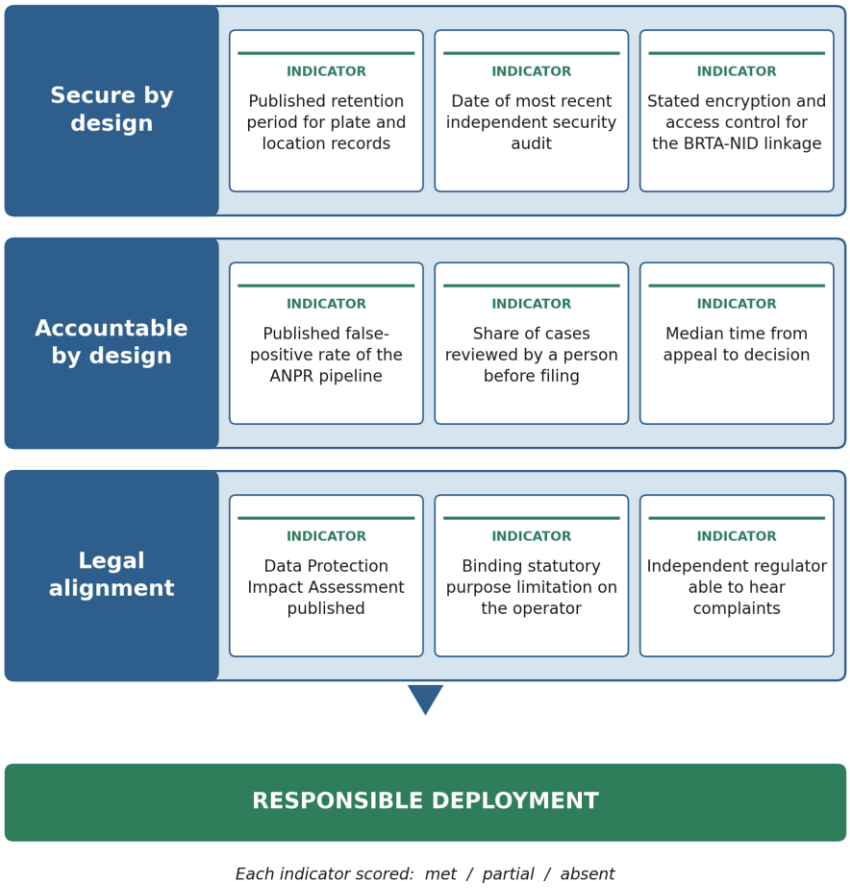


Fig. 2. A three-layer framework for responsible AI traffic surveillance, showing three observable indicators for each principle; Table II applies them to Dhaka. (Schematic rendering assisted by AI generation).

Fig. 2 sets out the three principles with their indicators; Table II turns them into a scoring instrument. Each indicator is scored as met, partial, or absent, so that a deployment can be rated rather than only described. The final column applies the instrument to Dhaka, where each entry records what the corpus of Section II does and does not contain.

Table II. Observable indicators for scoring a city against the three principles.

Principle	Observable indicator	Verified from	Dhaka status
Secure by design	Published retention period for plate and location records	Operator policy or public notice	None in corpus
	Date of most recent independent security audit	Audit summary from an external body	None in corpus
	Stated encryption and access control for the BRTA-NID linkage	Technical or procurement specification	Not stated
Accountable by design	Published false-positive rate of the ANPR pipeline	Operator transparency report	None in corpus
	Share of cases reviewed by a person before filing	Operator statistics or audit sample	Reported as none [3]
	Median time from appeal to decision	Published appeal statistics	No published pathway [1]
Legal alignment	Data Protection Impact Assessment published	Public register of assessments	None in corpus

	Binding statutory purpose limitation on the operator	Text of the enabling instrument	Broad state exemptions [16]
	Independent regulator able to hear complaints	Regulator establishment and casework	Expected about 2027 [16]

E. From Dhaka to Chattogram: Building It In Before It Hardens

Bangladesh's second city offers a natural test of this paper's central claim. Where Dhaka's system was built enforcement-first and leaves rickshaws outside its scope, Chattogram's planned system, still at the proposal stage, is designed around adaptive signal control that responds to real-time vehicle density and is intended to include rickshaws, vans and bicycles across fifty-three junctions, as reported by [4]. The contrast, summarized in Table III and placed on a timeline in Fig. 3, matters because it shows that the limitations of Dhaka's deployment were design choices, not technical inevitabilities. A city that has not yet built its system can still embed inclusive coverage, human oversight, and accountability from the outset, whereas retrofitting them into Dhaka's running system will be harder and costlier. The window for responsible design is widest before deployment, and Chattogram is currently inside it.

Table III. Two designs compared: Dhaka's deployed system and Chattogram's planned system.

Dimension	**Dhaka (deployed, May 2026)**	**Chattogram (planned, 2026)**
Primary function	Violation detection and automated enforcement	Adaptive signal control for traffic flow
Vehicle coverage	Motor vehicles; rickshaws excluded	Includes rickshaws, vans and bicycles
Scale	About 25 intersections	53 junctions planned
Human role	Fully automated case filing	Flow optimization; enforcement design still open
Design opportunity	Safeguards must be retrofitted	Safeguards can be built in from the start

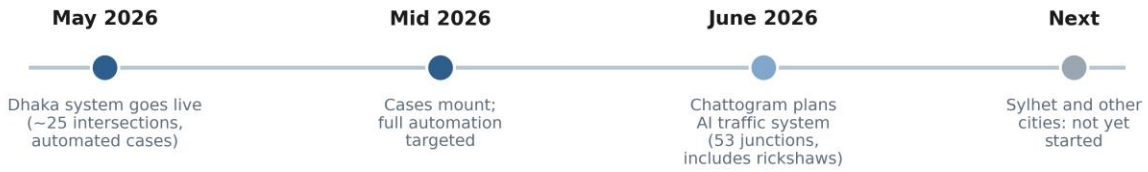


Fig. 3. Timeline of AI traffic surveillance across Bangladeshi cities, from Dhaka's 2026 deployment to Chattogram's planned system and cities yet to begin. (Chronological graphic rendering assisted by AI generation; milestone data compiled from case study sources).

IV. CONCLUSION

AI traffic enforcement is useful and, in Bangladesh, here to stay. But a system that sees every plate and links it to an identity is surveillance infrastructure, and unchecked surveillance is dangerous. The argument of this paper is that responsible governance is the precondition for a genuinely better tomorrow: safer roads and protected citizens, not one purchased at the cost of the other. The country has moved quickly on the technology; it now needs to move with equal speed on security and accountability, embedding both by design as the system scales from Dhaka to Chattogram, Sylhet, and beyond.

This study has limitations. As Section II notes, no public technical documentation or independent audit exists, which bounds the precision of any external assessment. The analysis is conceptual rather than empirical, the regulatory regime is still immature, and safeguards depend on institutional and political commitment this paper cannot guarantee. These limits, however, reinforce rather than weaken its central claim: where transparency is scarce, the case for building accountability in becomes stronger, not weaker.

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